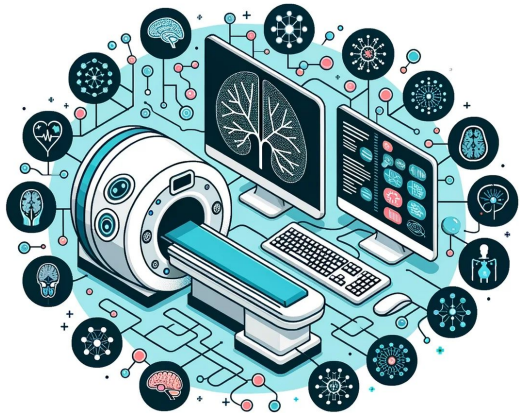




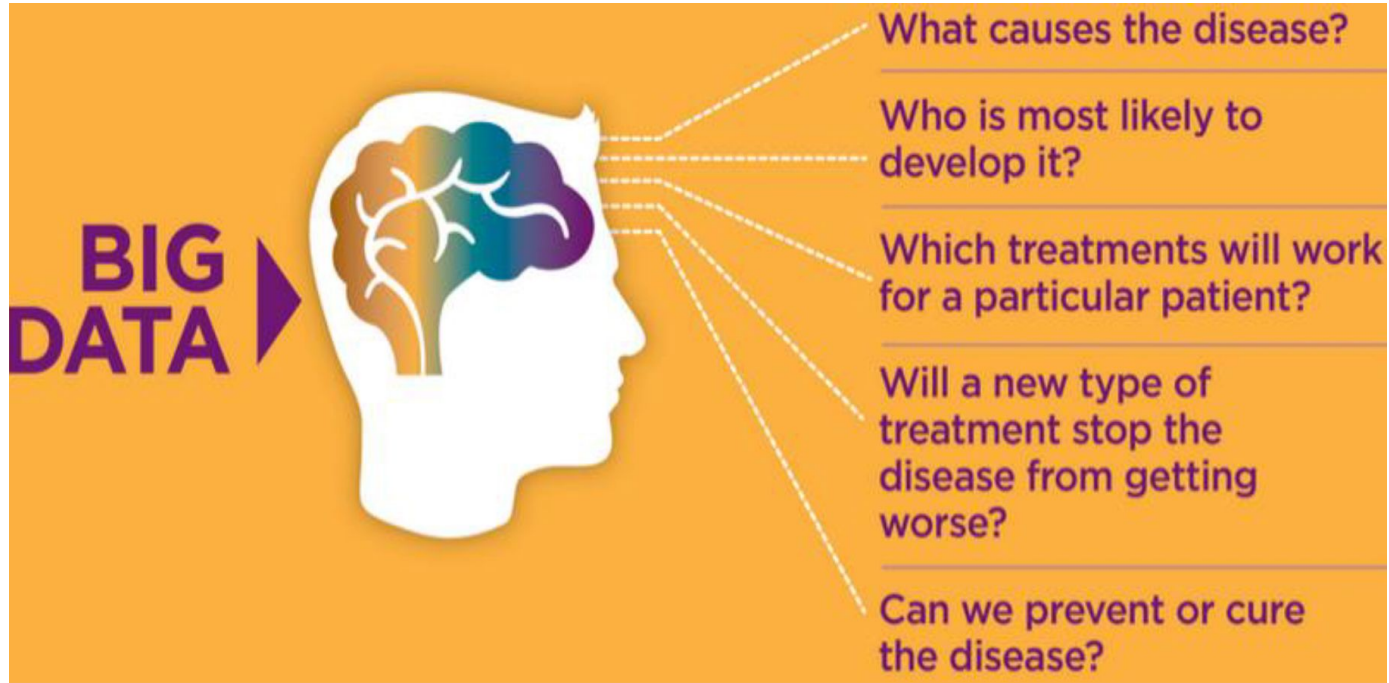
Medical Image Segmentation and Classification

Session 07

Oct 18 2025

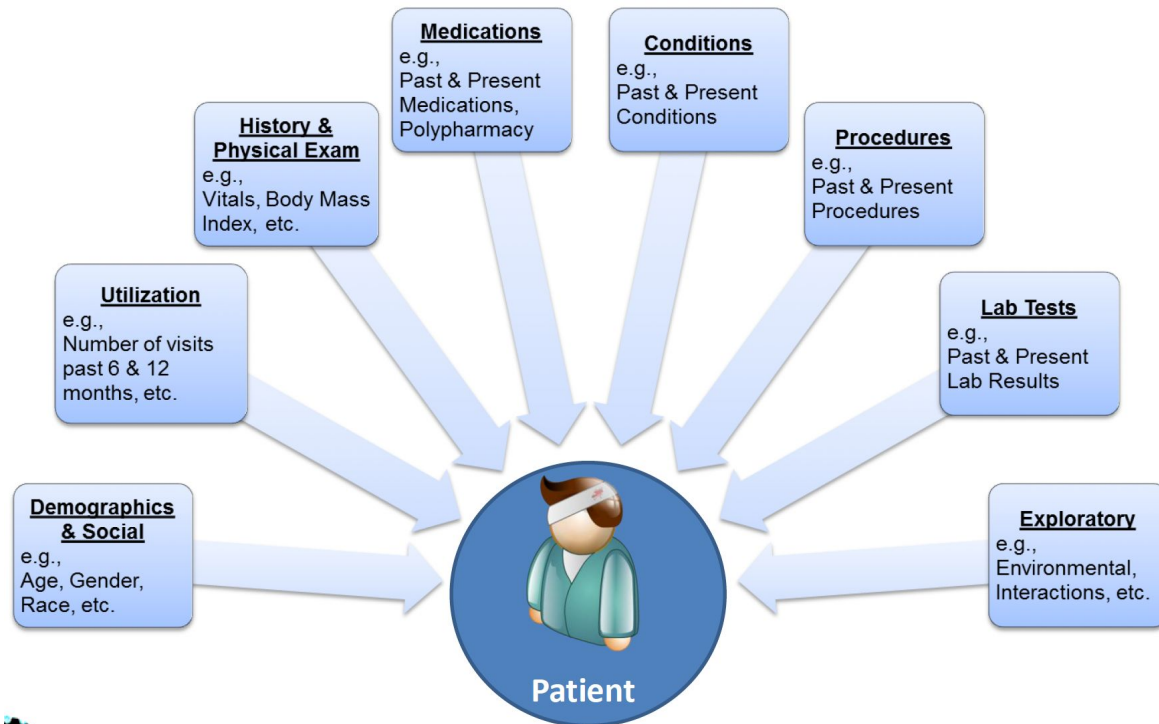
Big data in healthcare

- There are so many **fundamental questions** to answer!



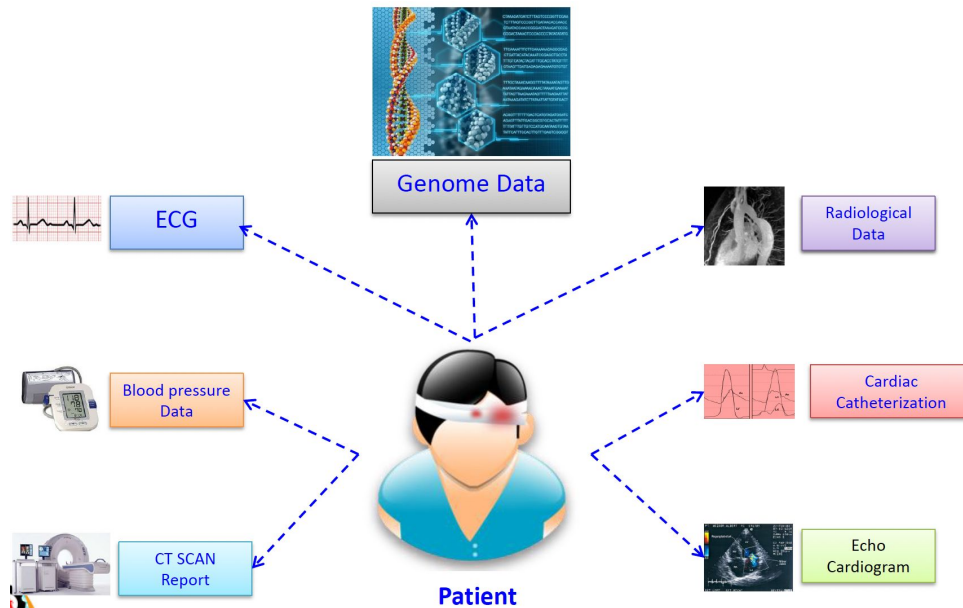
Structured medical data

- Data collection **does not require imaging**

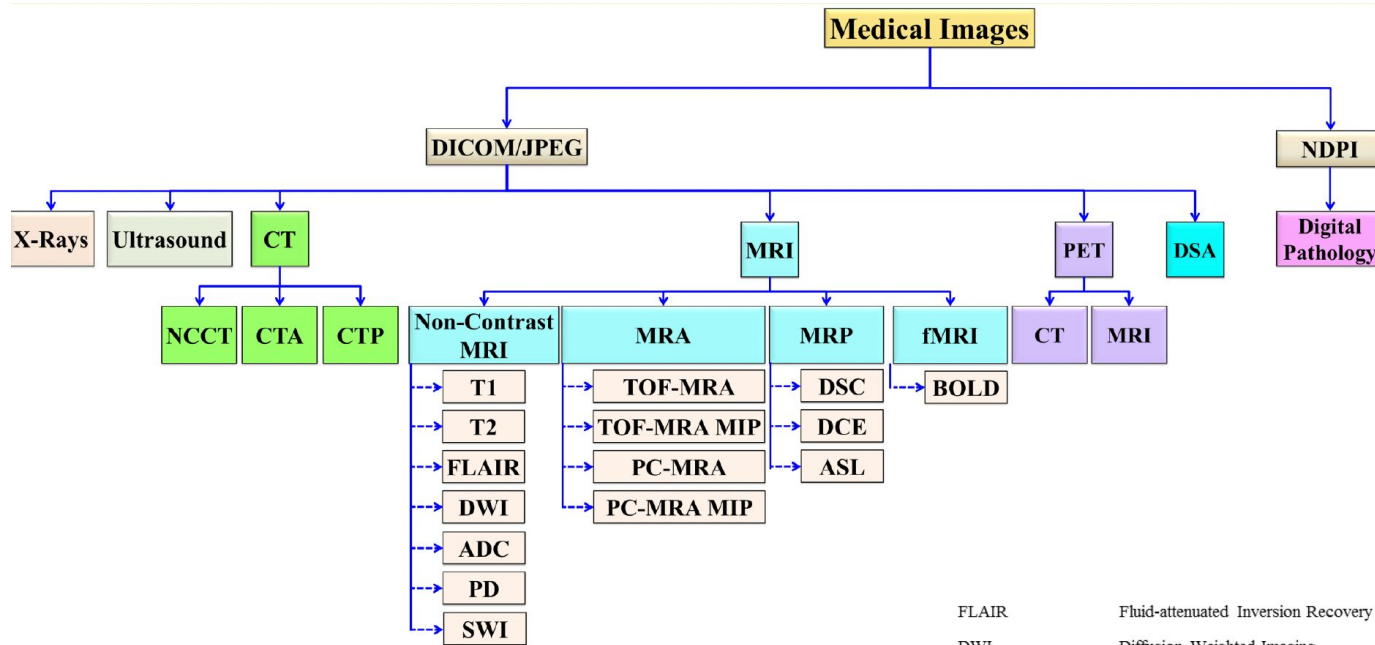


Unstructured multimodal clinical data

- ECG, CT, MRI, genomic sequences, etc.
 - typically grouped under **medical imaging and signals** or **unstructured clinical data**.



Types of medical images



CT Computed Tomography
 NCCT Non-contrast CT
 CTA CT Angiography
 CTP CT Perfusion
 MRI Magnetic Resonance Imaging

MRP Magnetic Resonance Perfusion
 DSC Dynamic Susceptibility Contrast
 DCE Dynamic Contrast Enhanced
 ASL Arterial Spin Labelling
 BOLD Blood-oxygen-level Dependent
 PET Positron Emission Tomography
 DSA Digital Subtraction Angiography

FLAIR Fluid-attenuated Inversion Recovery
 DWI Diffusion Weighted Imaging
 ADC Apparent Diffusion Coefficient
 PD Proton Density
 SWI Susceptibility Weighted Imaging
 MRA Magnetic Resonance Angiography
 TOF-MRA Time-of-Flight MRA
 TOF-MRA MIP TOF-MRA Maximum Intensity
 PC-MRA Phase-contrast MRA

Deep learning is here to help!

- Using sound to diagnose tuberculosis



Deep learning is here to help!

LETTERS

<https://doi.org/10.1038/s41591-019-0447-x>

nature
medicine

Corrected: Author Correction

End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography

Diego Ardila ^{1,5}, Atilla P. Kiraly^{1,5}, Sujeeth Bharadwaj^{1,5}, Bokyung Choi^{1,5}, Joshua J. Reicher², Lily Peng¹, Daniel Tse ^{1*}, Mozziyar Etemadi ³, Wenxing Ye¹, Greg Corrado¹, David P. Naidich⁴ and Shravya Shetty¹

Deep learning is here to help!

ARTICLES

<https://doi.org/10.1038/s41551-019-0487-z>

nature
biomedical engineering

There are amendments to this paper

Detection of anaemia from retinal fundus images via deep learning

Akinori Mitani ^{1*}, Abigail Huang¹, Subhashini Venugopalan ², Greg S. Corrado¹, Lily Peng¹, Dale R. Webster ¹, Naama Hammel¹, Yun Liu ^{1,3} and Avinash V. Varadarajan^{1,3}

Owing to the invasiveness of diagnostic tests for anaemia and the costs associated with screening for it, the condition is often undetected. Here, we show that anaemia can be detected via machine-learning algorithms trained using retinal fundus images, study participant metadata (including race or ethnicity, age, sex and blood pressure) or the combination of both data types (images and study participant metadata). In a validation dataset of 11,388 study participants from the UK Biobank, the meta-data-only, fundus-image-only and combined models predicted haemoglobin concentration (in g dL⁻¹) with mean absolute error values of 0.73 (95% confidence interval: 0.72–0.74), 0.67 (0.66–0.68) and 0.63 (0.62–0.64), respectively, and with areas under the receiver operating characteristic curve (AUC) values of 0.74 (0.71–0.76), 0.87 (0.85–0.89) and 0.88 (0.86–0.89), respectively. For 539 study participants with self-reported diabetes, the combined model predicted haemoglobin concentration with a mean absolute error of 0.73 (0.68–0.78) and anaemia an AUC of 0.89 (0.85–0.93). Automated anaemia screening on the basis of fundus images could particularly aid patients with diabetes undergoing regular retinal imaging and for whom anaemia can increase morbidity and mortality risks.

Preparing data for automated diagnosis

- Importance of **standardizing datasets** (e.g., LDCT scans, ECG, EEG, MRI, Genomic Data).
 - Data preprocessing techniques
 - **Normalization & Augmentation**: Improving model robustness.
 - **Segmentation Models (U-Net, nnU-Net)**: For accurate ROI identification.
 - **Data Cleaning & Annotation**: Handling missing data, noise, and bias.
-

Building and training deep learning models

- Common architectures
 - **CNNs**: for image-based diagnosis
 - **RNNs**: for sequential data like ECG/EEG
 - **Transformer models**: Handling complex data patterns (genomics)
 - Feature extraction techniques
 - **3D convolutions**: For volumetric analysis (e.g., MRI, CT)
 - **Hybrid models**: Combining multiple architectures
 - Training best practices
 - **Transfer learning**: Using pre-trained models
 - **Cross-validation and hyperparameter tuning**: Enhancing model performance
-

Evaluation and model validation

- Key metrics for diagnosis models
 - **Accuracy, Precision, Recall, F1-Score**
 - Classification performance
 - **AUROC (Area under the receiver operating characteristic curve)**
 - General performance
 - **Dice coefficient, IoU (intersection over union):**
 - Segmentation quality
 - Model validation
 - **Cross-validation:** Ensuring robustness
 - **External validation:** Using independent datasets.
-

Confusion matrix

		PREDICTED LABEL	
		NEGATIVE	POSITIVE
TRUE LABEL	NEGATIVE	TRUE NEGATIVE	FALSE POSITIVE
	POSITIVE	FALSE NEGATIVE	TRUE POSITIVE

		PREDICTED LABEL	
		NEGATIVE	POSITIVE
TRUE LABEL	NEGATIVE	 TRUE NEGATIVE	 FALSE POSITIVE
	POSITIVE	 FALSE NEGATIVE	 TRUE POSITIVE

Classification evaluation metrics

Sources: [4][5][6][7][8][9][10][11] view · talk · edit

		Predicted condition			
		Predicted positive	Predicted negative	Informedness, bookmaker informedness (BM) $= \text{TPR} + \text{TNR} - 1$	Prevalence threshold (PT) $= \frac{\sqrt{\text{TPR} \times \text{FPR}} - \text{FPR}}{\text{TPR} - \text{FPR}}$
Actual condition	Total population $= P + N$				
	Positive (P) [a]	True positive (TP), hit ^[b]	False negative (FN), miss, underestimation	True positive rate (TPR), recall, sensitivity (SEN), probability of detection, hit rate, power $= \frac{\text{TP}}{P} = 1 - \text{FNR}$	False negative rate (FNR), miss rate type II error ^[c] $= \frac{\text{FN}}{P} = 1 - \text{TPR}$
	Negative (N) ^[d]	False positive (FP), false alarm, overestimation	True negative (TN), correct rejection ^[e]	False positive rate (FPR), probability of false alarm, fall-out type I error ^[f] $= \frac{\text{FP}}{N} = 1 - \text{TNR}$	True negative rate (TNR), specificity (SPC), selectivity $= \frac{\text{TN}}{N} = 1 - \text{FPR}$
		Positive predictive value (PPV), precision $= \frac{\text{TP}}{\text{TP} + \text{FP}} = 1 - \text{FDR}$	False omission rate (FOR) $= \frac{\text{FN}}{\text{TN} + \text{FN}} = 1 - \text{NPV}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$
		False discovery rate (FDR) $= \frac{\text{FP}}{\text{TP} + \text{FP}} = 1 - \text{PPV}$	Negative predictive value (NPV) $= \frac{\text{TN}}{\text{TN} + \text{FN}} = 1 - \text{FOR}$	Markedness (MK), deltaP (Δp) $= \text{PPV} + \text{NPV} - 1$	Diagnostic odds ratio (DOR) $= \frac{\text{LR}^+}{\text{LR}^-}$
		F_1 score $= \frac{2 \text{PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}} = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}$	Fowlkes–Mallows index (FM) $= \sqrt{\text{PPV} \times \text{TPR}}$	Matthews correlation coefficient (MCC) $= \frac{\text{TPR} \times \text{TNR} \times \text{PPV} \times \text{NPV}}{\sqrt{\text{FNR} \times \text{FPR} \times \text{FOR} \times \text{FDR}}}$	Threat score (TS), critical success index (CSI), Jaccard index $= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}$

End-to-end lung cancer screening

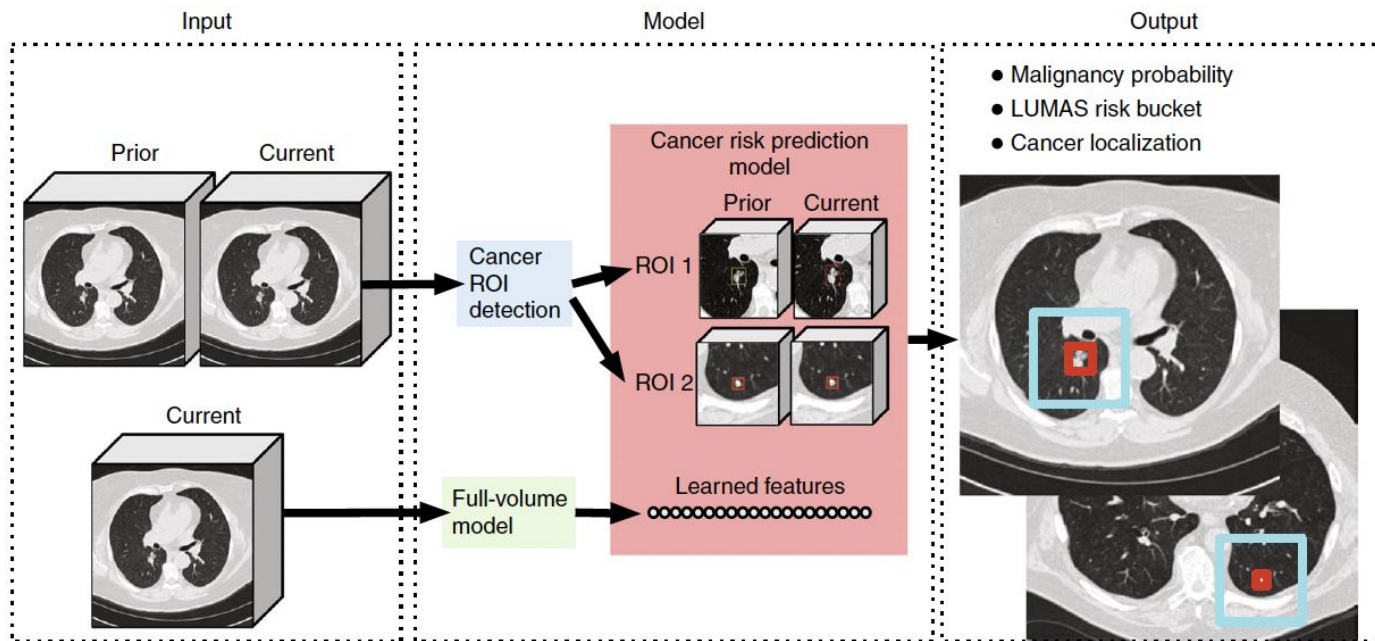
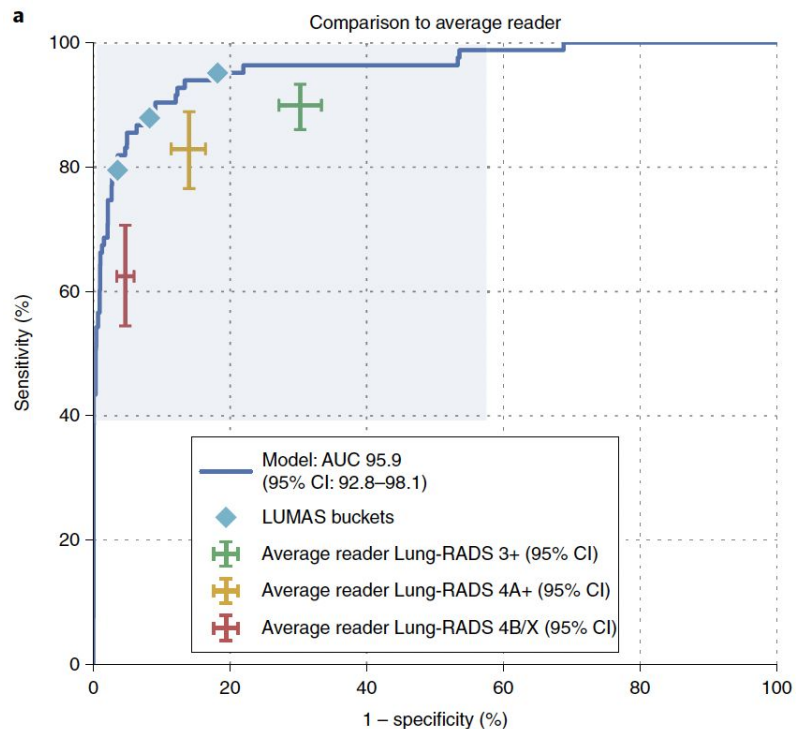


Fig. 1 | Overall modeling framework. For each patient, the model uses a primary LDCT volume and, if available, a prior LDCT volume as input. The model then analyzes suspicious and volumetric ROIs as well as the whole-LDCT volume and outputs an overall malignancy prediction for the case, a risk bucket score (LUMAS) and localization for predicted cancerous nodules.

End-to-end lung cancer screening

- multi-step process, involving segmentation, detection, feature extraction, and prediction.
 - Preprocessing
 - Converting CT scans to standardized input format.
 - Segmentation & ROI Detection
 - Using models to locate suspicious regions (nodules).
 - Feature Extraction & Diagnosis
 - Feeding ROIs to a cancer risk prediction model.
 - Output
 - Generating malignancy probabilities, LUMAS risk scores, and nodule localization.
-

Lung cancer screening results



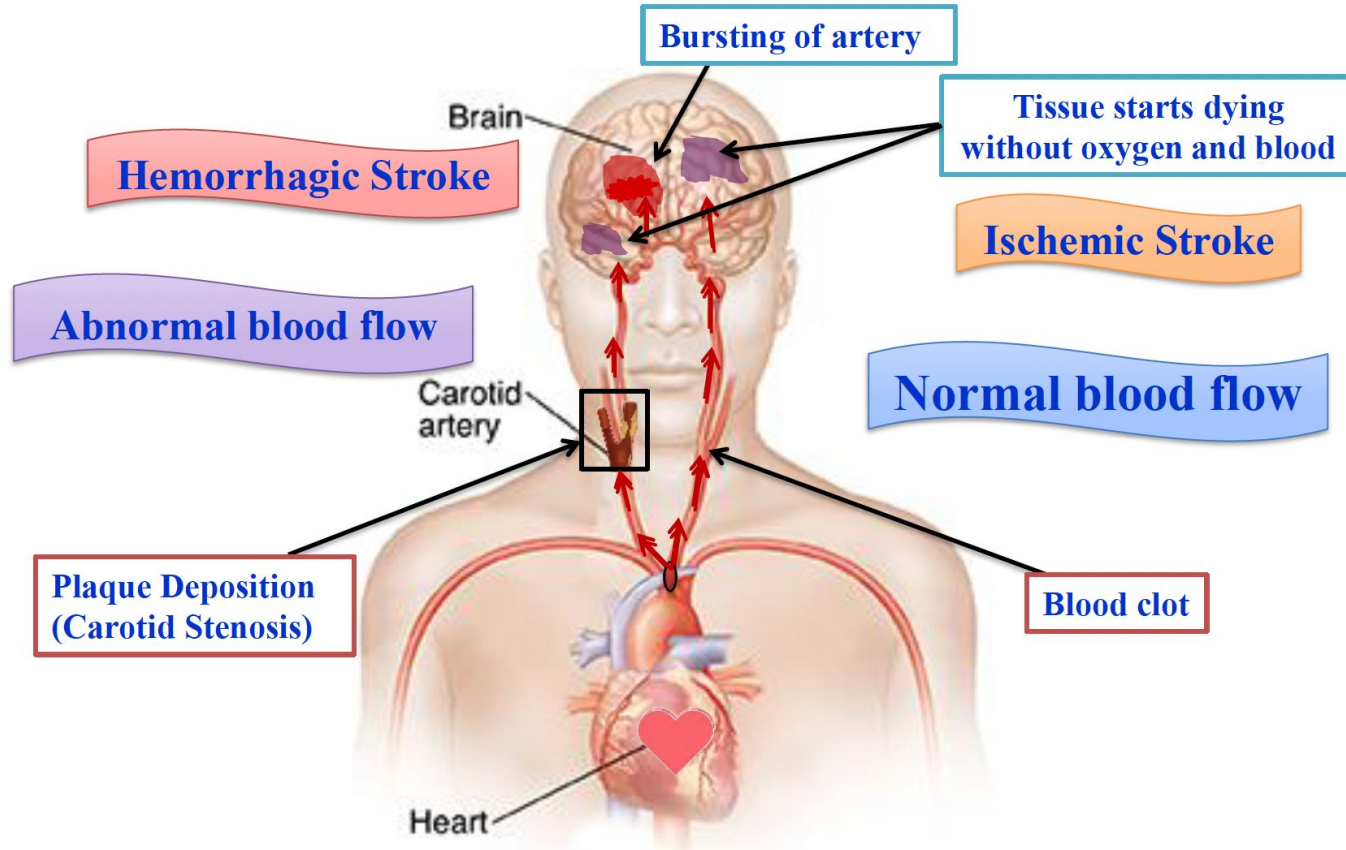
c

Risk buckets		Sensitivity	Delta
1,2 versus 3+	Average reader	90.0 (86.1, 93.4)	+5.2*
	Model	95.2 (89.9, 98.9)	(.4, 9.8) $P = 0.0386$
1,2,3 versus 4A+	Average reader	82.9 (76.6, 89.4)	+7.4*
	Model	90.4 (83.3, 96.3)	(1.7, 12.9) $P = 0.0114$
1,2,3,4A versus 4B/X	Average reader	62.5 (54.4, 70.7)	+17.1*
	Model	79.5 (70.8, 88.2)	(9.0, 24.5) $P < 1 \times 10^{-4}$

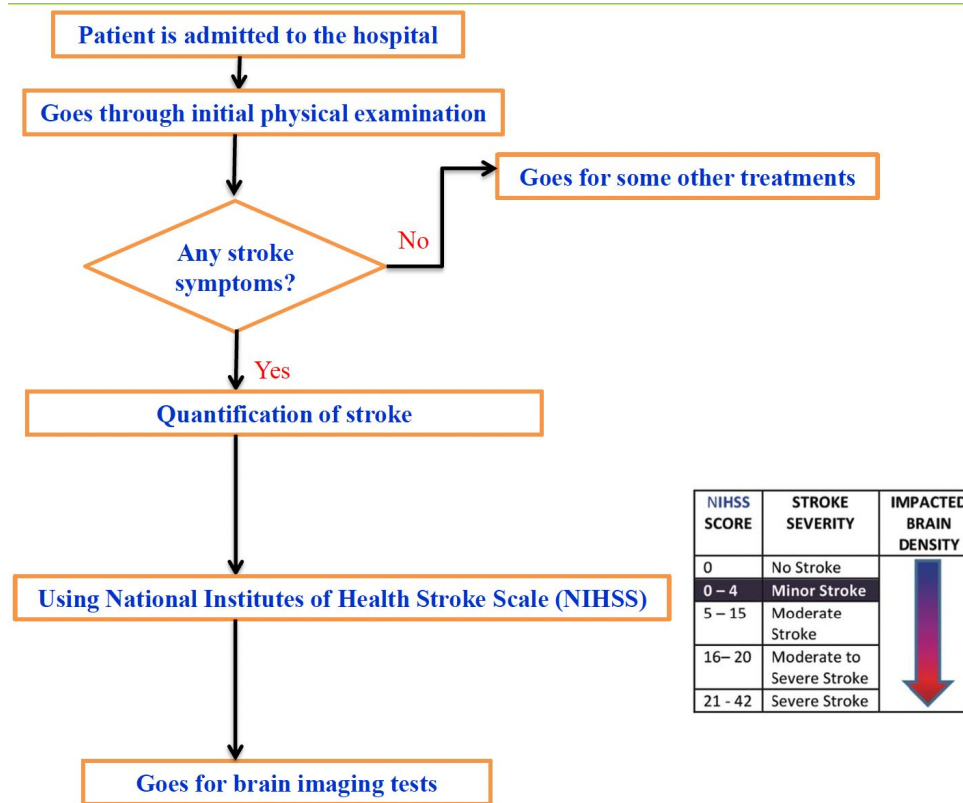
d

Risk buckets		Specificity	Delta
1,2 versus 3+	Average reader	69.7 (66.6, 72.8)	+11.6*
	Model	81.3 (77.3, 84.9)	(7.8, 15.1) $P < 1 \times 10^{-4}$
1,2,3 versus 4A+	Average reader	86.0 (83.4, 88.4)	+5.0*
	Model	91.0 (88.1, 93.9)	(1.7, 12.9) $P = 0.0008$
1,2,3,4A versus 4B/X	Average reader	95.3 (94.0, 96.6)	+1.1*
	Model	96.5 (94.6, 98.2)	(-0.4, 2.6) $P = 0.143$

Cerebrovascular disease

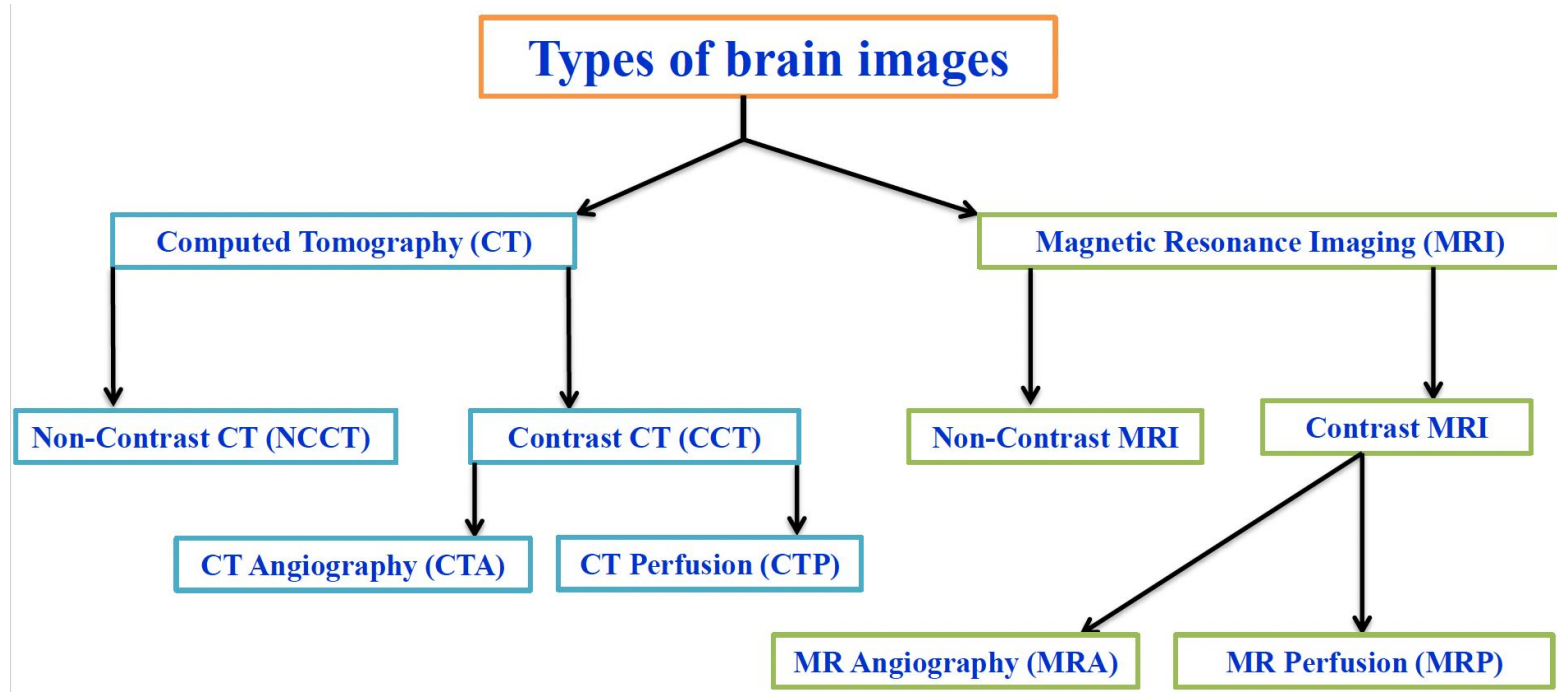


Diagnosis methods



Types of brain images

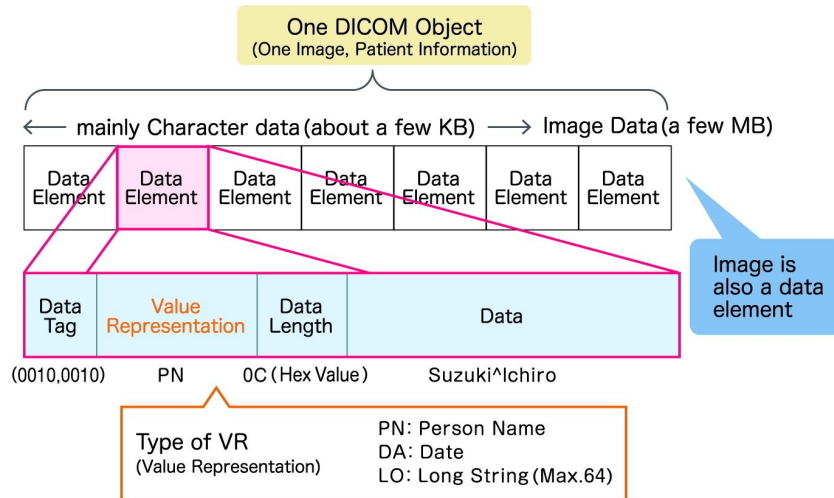
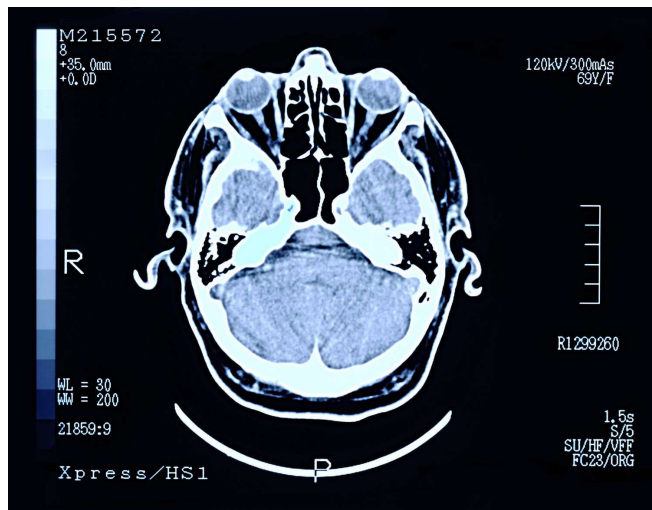
- External iodine material used for contrast



Common brain image format

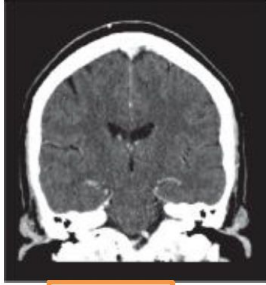
CT/MRI

DICOM (Digital Imaging and Communications in Medicine)

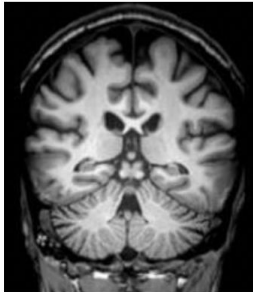


Imaging planes

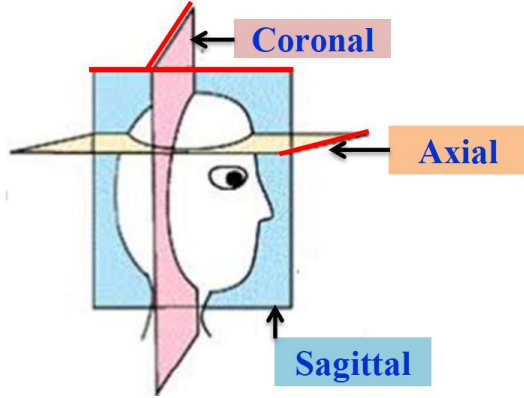
Coronal
(front and back view)



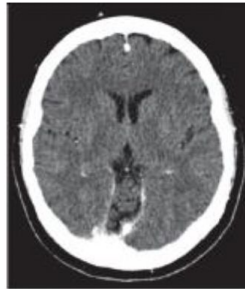
NCCT



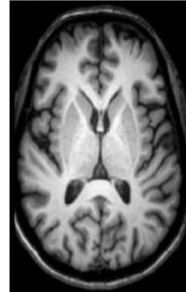
MRI



Axial (horizontal view)

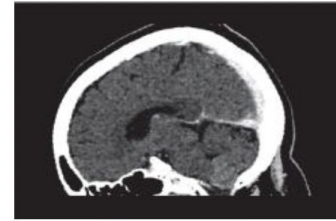


NCCT



MRI

Sagittal (lateral view)



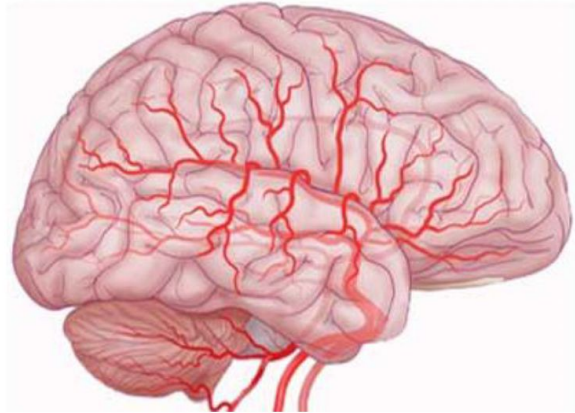
NCCT



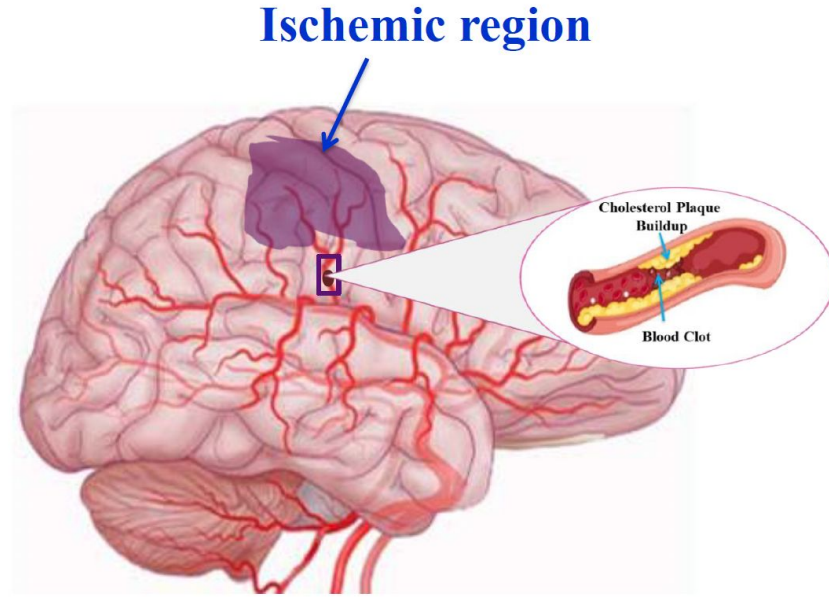
MRI

Brain stroke

- An **ischemic brain** refers to a brain that has suffered from **ischemia**, which is a condition where there is a lack of blood flow to a part of the brain, causing oxygen and nutrient deprivation.

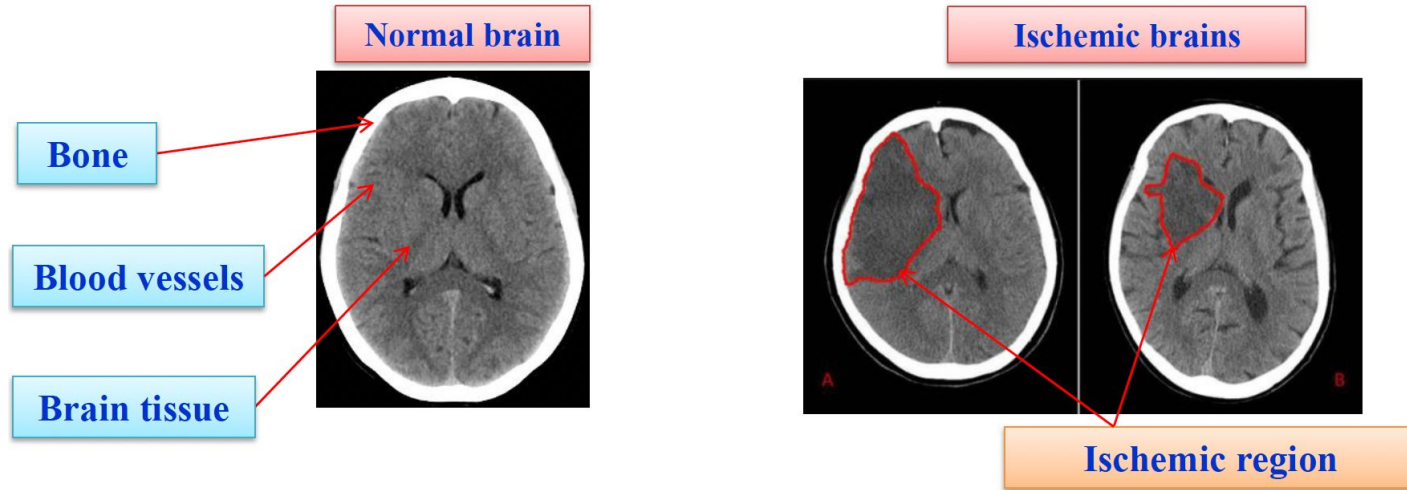


Normal Human Brain



Brain of Ischemic Stroke Patient

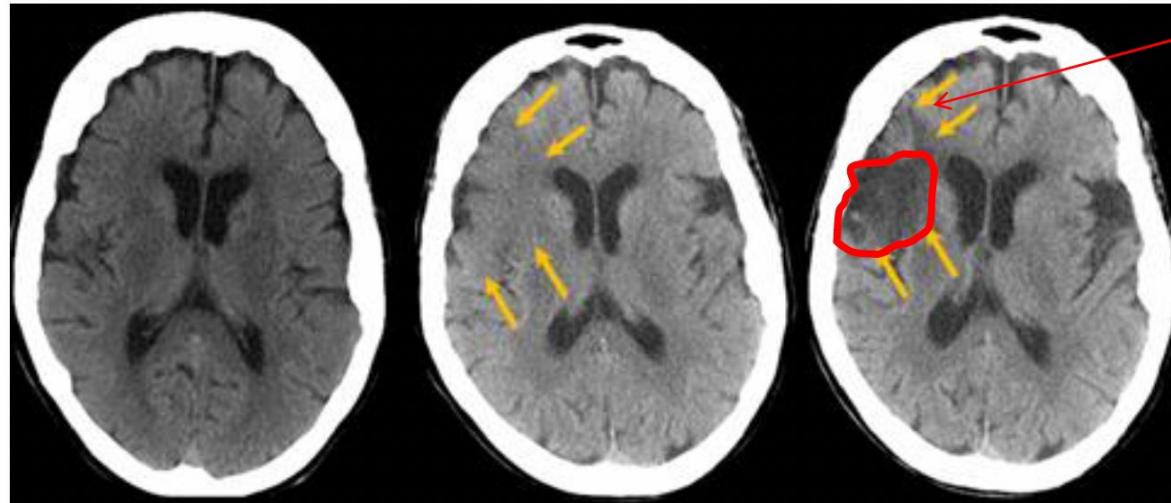
Image of normal vs. ischemic brain



Color difference between the parts of normal vs. ischemic brain		
Parts	Colors	
	Normal brain	Ischemic brain
Bone	White	White
Blood vessel	Mostly white	Mostly white
Brain tissue	Mostly gray	Dark gray (Ischemic region)

Normal progression of infarction

- Infarction
 - Permanent tissue death from prolonged ischemia (irreversible)
 - Difficult to find out in brain CT taken within hours of stroke!



Ischemic region
(the brain tissue
color is changed
to dark grey)

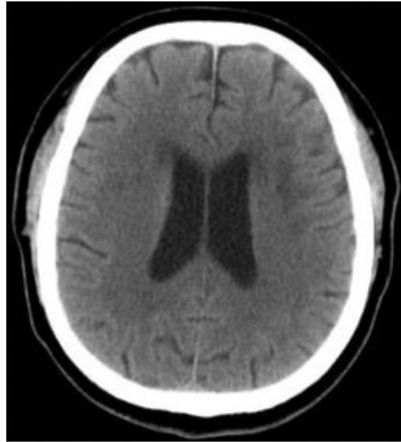
3 hours

1 day

5 days

Clinical difficulties

- **Difficult to distinguish** between normal and ischemic stroke patients with naked eye
 - **Similar color intensity** of brain tissue



NCCT image: Normal patient



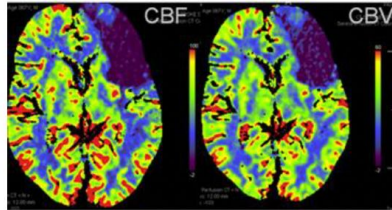
NCCT image: Ischemic stroke patient

Clinical difficulties

- CTA or CTP
 - injecting a **contrast agent (dye)** into the bloodstream to visualize blood vessels in the brain and neck.
 - Uses high rates of iodine-rich contrast material
 - Carcinogenic
 - Potential of allergic reactions
 - Health issues corresponding to radiation does



CTA

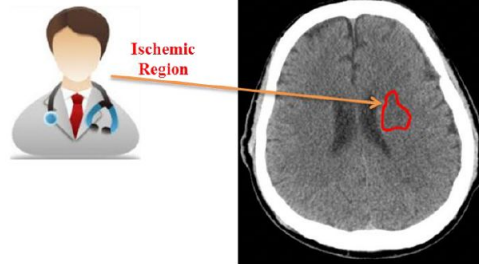
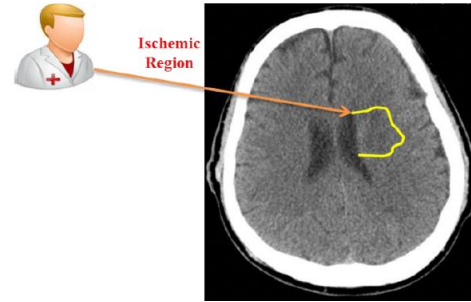
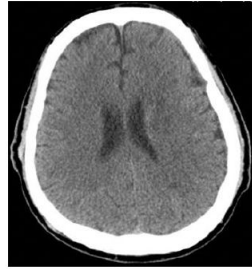


CTP



Technical difficulties

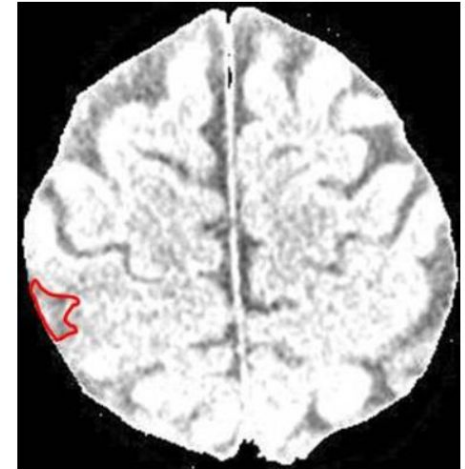
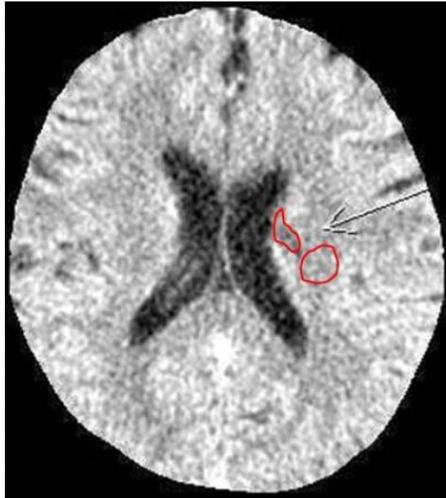
- Localization of ischemic region is difficult!



NCCT image: Ischemic stroke patient

Technical difficulties

- Ischemic stroke has no particular location!



NCCT images: Ischemic stroke patients

Ischemic stroke diagnosis using AI

- Classification of **normal** and **ischemic** stroke patients with deep learning
- **Localization** of ischemic stroke region with deep learning
- Models to use
 - AlexNet, VGGNet, GoogleNet, Inception, ResNet, etc.
 - Transfer learning
 - Use the models pretrained on a large corpus of nature images
 - Finetune it on the existing labeled data

Training data

- The input data are NCCT images
- Ground truth are determined using DWI

